

The exact path of `netbeans.conf` for the various OS can be found here: <http://wiki.netbeans.org/FaqNetbeansConf>

Making NetBeans project

We'll try to set up project for Microstrain 3DM-GX2 IMU driver package, so note it's path:

- `1 $ roscd microstrain_3dmgx2_imu`
- `2 $ pwd`
- Start "C/C++ Project with Existing Sources" wizard
- Use the above path to "Specify the folder that contains existing sources"
- Select "Custom" Configuration Mode (note that Automatic Mode recognizes `cmake`)
- Proceed with the wizard accepting defaults

You should get a NetBeans project for `microstrain_3dmgx2_imu` package with automatically configured Code Assistance (and almost working for all dependencies). E.g. you can see that the bullet library headers weren't parsed.

We will configure Code Assistance manually. That means entering paths to all header files the package uses and all preprocessor definitions.

To get the paths to include files for the package we will use `rospack`. Further, we'll use `sed` to format them for easier input to NetBeans.

- `1 $ rospack cflags-only-I microstrain_3dmgx2_imu | sed 's/ /:/g'`

Open the project properties. Go to Code Assistance -> C++ Compiler and paste the output of the above command to Include Directories field.

Use `rospack` to find the preprocessor definitions and enter them manually.

- `1 $ rospack cflags-only-other microstrain_3dmgx2_imu`

Code auto-formatting in NetBeans

The following file `netbeans-ros-code_style.zip` is prepared to enable auto-formatting of C++ code in NetBeans as defined in `CppStyleGuide`. In order to use it, you should import it to Netbeans (Tools -> Options -> Import).

With this, the example given in `CppStyleGuide#Formatting` will be identically formatted, except for extra blank lines before function or class definitions. For a discussion see Google C++ style guide Vertical Whitespace.

2. PYCHARM

PyCharm is an IDE for Python. In order to run and debug ROS functionality,